

The one API to rule them all

When it comes to software that helps you build AI models, we have a wide variety of libraries, APIs and frameworks to choose from. There are libraries which were built for a certain purpose and then there are general purpose libraries. Some libraries even get some much needed love from hardware manufacturers and we see optimisations at the hardware level to help accelerate these libraries further. After all, the vast majority of AI libraries rely on highly parallelised computing architecture to speed up the processing. With so many different APIs, libraries and frameworks, the problem lies with figuring out which to learn and given that the AI domain is nascent, you really can't spend that much of time moving from one library to another till you find the one that suits your needs best.

This is where Keras comes into the picture. Keras is a high-level neural network API written in Python which means it runs on top of an AI library such as TensorFlow, Theano or CNTK. It would be better if you avoid Theano since they've ceased development and focus instead on TensorFlow and CNTK. The key feature of Keras is that this compatibility allows you to switch out the underlying library without performing any code changes to the backend i.e. the library its running on. One day you could have your model running on TensorFlow and then switch to CNTK the very next day with ease. Essentially, Keras is democratising AI by reducing the barrier for entry into the AI / ML world. So much that, it seems that if you wish to get into data science, then Keras should be your first step.